Smith's explicit isogeny ψ on divisor classes (W3b), and a toy Route S solve

`research/smith_isogeny_divisor_map_20261018` · branch `claude/smith-isogeny-divisor-map-20261018` · protocol `PROTOCOL.md` (frozen 2026-10-18, not edited) · synthetic toy fields only

1. Result in one paragraph

$\psi = (\pi_X)_*(\pi_H)^*$ (Smith 2009, arXiv:0806.2995, §§5–6) was computed on Mumford divisors of genus-3 curves H: y² = F(x), deg F = 7, as effective degree-6 divisors on the plane quartic C ≅ X that the same map's canonical image defines, and classes were compared with two independent engines: W1's canonical `Pic^0` keys (`plane_quartic_pic::PlaneQuartic::reduce_deg6`, `PicClass::key`) and the branch's own degree-6 adjoint reduction. **No check failed on either engine** (pass/fail/skip below; a skip is an instance the engine could not decide and is not counted as a pass). On every curve at q ≤ 13 the full image was enumerated: $\ker \psi \cap J_H(F_q) = S(F_q)$ and every fibre has $\#S(F_q)$ elements. Route S ran end to end at toy q: a planted ECDLP on $E0/F_{q^3}$ went through the Joux–Vitse m = 3, r = 5 descent to H, through ψ to $J_X(F_q)$, was solved there on canonical keys, and the recovered scalar was checked by scalar multiplication on E0. No security claim about any curve is made or implied.

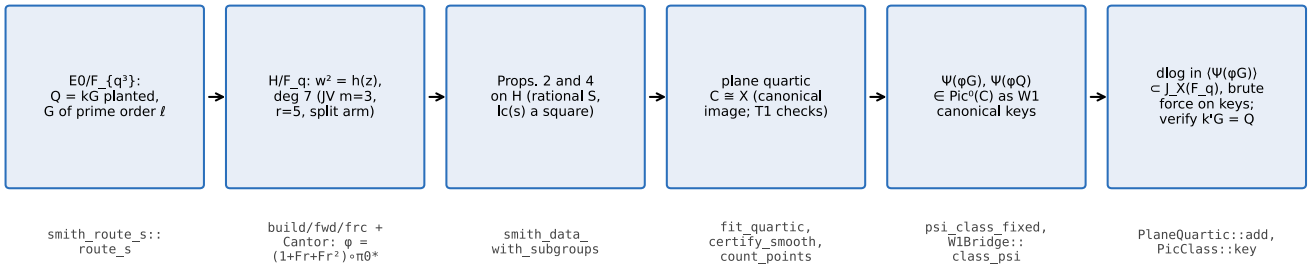


Figure 1: Route S as executed (T8). Module and function names under each box.

2. Instances and what was run

Master seed 20261018 (SplitMix64). Verification suite: q ∈ {7, 11, 13, 17, 19, 23}; per q the first 6 monic squarefree degree-7 F (stream `random_curve7`, the `random_curve` scheme of `smith_prop24` with degree 7) having a rational (S, g) (Props. 2 and 4); the first such map is used. Sample per curve (protocol §4): 30 T2 elements with two bases A, A′; 30 T3 pairs; all nonzero elements of $S(F_q)$ and 100 random elements outside it (T4); 3 elements of order ℓ for each odd prime ℓ | $\#J_H(F_q)$ (T5); T6 on the T2 images. Engines see identical ψ-images.

q	curves scanned	curves used	T1 pass	X hyperelliptic (T1a)	hold-out pts / fail	image enumerated (curves, elements)	image = J/S, ker = S	first-base retries
7	33	6	6	0	346 / 0	6, 2758	6 / 6	136
11	36	6	6	0	377 / 0	6, 10020	6 / 6	167
13	45	6	6	0	397 / 0	6, 13342	6 / 6	139
17	40	6	4	2	278 / 0	0, 0	0 / 0	0
19	45	6	6	0	414 / 0	0, 0	0 / 0	0
23	22	6	4	2	284 / 0	0, 0	0 / 0	0

T1 passes means: quartic nullspace of dimension 1, hold-out of fresh image points (all on C), smoothness certificate (two resultant frames) and W1's own certificate, $\#C(F_{q^k}) = \#H(F_{q^k})$ for k = 1, 2, 3 and equal L-polynomials. "X hyperelliptic" is T1(a)'s separately reported outcome (images on a conic): no plane quartic, so T2–T6 do not apply to that curve. "First-base retries": elements whose image under the first base had a point of multiplicity > 2 (which the bridge does not reduce) and were recomputed with another base (the class does not depend on it, T2).

3. T2–T6: two engines side by side (pass / fail / skip, all q)

check	W1 canonical keys	degree-6 adjoint engine
T2 well-defined	932 / 0 / 28	793 / 0 / 167
T3 homomorphism	917 / 0 / 43	778 / 0 / 182
T4 S ⊂ ker	74 / 0 / 0	68 / 0 / 6
T4 ≠ 0 off S	3147 / 0 / 53	2927 / 0 / 273
T5 Ψ(G) ≠ 0	177 / 0 / 3	164 / 0 / 16
T5 ℓΨ(G) ~ 0	177 / 0 / 3	126 / 0 / 54
T6 #J·Ψ ~ 0	932 / 0 / 0	516 / 0 / 277

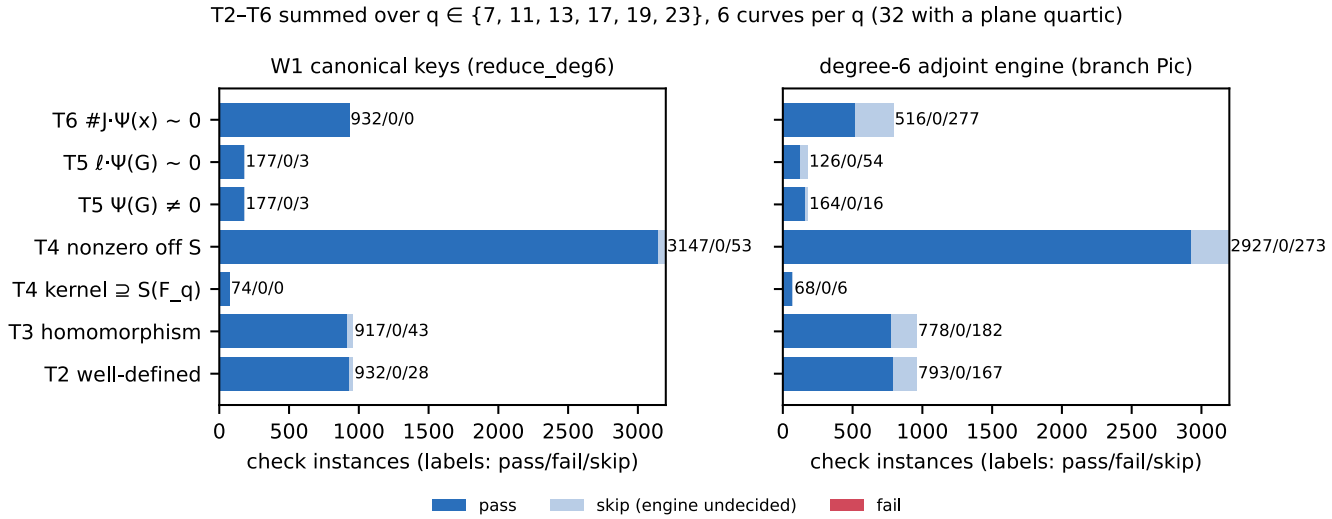


Figure 2: T2–T6 instances by outcome, summed over q ∈ {7, …, 23} (per-q tallies in `data/summary.json`, per-curve rows in `data/verify_q*.jsonl`).

W1 skips are images with a point of multiplicity > 2 (the bridge builds the cubics through Z_6 with tangency rows up to multiplicity 2) and fall with q. Degree-6 engine skips are its own exceptional/overlap/frame failures (listed per curve in the JSONL); its T6 skips are scalar multiplications that hit such a case. The degree-6 engine was validated by its group-law test at p = 37 (associativity, commutativity, inverses, #J·a = 0) and by agreeing with W1 on every instance both decided; the protocol's T9(ii) class count at q = 7 for this engine was **not run**. Which arithmetic was kept: W1's `PlaneQuartic` is the arithmetic (group law, keys, the T8 discrete log); the branch's adjoint `Pic` stays only as the protocol's independent second engine.

T1 on degree-8 models and T1(f).

q	curves	T1 pass	X hyperelliptic	twist maps (lc(s) non-square)	$N_1(X) = 2(q+1) - N_1(H)$	not applicable / disagree
7	6	4	2	25	7	18 / 0
11	6	6	0	2	0	2 / 0
13	6	6	0	17	9	8 / 0
17	6	6	0	11	4	7 / 0
19	6	5	1	17	15	2 / 0
23	6	6	0	6	4	2 / 0

T1(f) is checked at k = 1 only (`smith_prop24::x_point_count` exists for deg F = 8, k = 1; "not applicable" = a degenerate fibre or $N_1(H) = q + 1$); k = 2, 3 of T1(f) were not run.

4. T8: Route S end to end

Construction: `research/ghs_cubic_hyperelliptic_genus3_20261013/hyper_genus3.rs` (Joux–Vitse m = 3, r = 5, split arm Q = (x − a)(x − b)), ported into `src/cryptanalysis/smith_route_s.rs`, candidates in its own order (census index of r_0, then Q). Protocol filter: #E0(F_{q^3}) has a prime factor $\ell \geq 5$ (the largest is used; the old prototype's #E0 = 4ℓ / 8ℓ quota rule is recorded beside it). Per instance: descend, make h monic, Props. 2 and 4, then (where they pass) the planted DLP: G of order ℓ, Q = kG (k from the seed), φ(G), φ(Q) by Cantor (homomorphism and order checked), C fitted and T1-checked, Ψ(φG), Ψ(φQ) as W1 keys, brute-force search of k′ in ⟨Ψ(φG)⟩, then k′G = Q on E0.

q	candidates	filter ($\ell \geq 5$)	run	not reached (1 h cap)	Props. 2+4 pass	DLP tried	X quartic (T1)	$\#J_H = \#E0$	$\Psi(\varphi G) \neq 0$, $\ell\Psi = 0$	k′G = Q verified	k′ = k
7	2352	2352	2352	0	1596	1596	1300	1300	1300	1300	1300
11	24200	22000	22000	0	18150	18150	15662	15662	15662	15662	15662
13	56784	56784	56784	0	44616	44616	40562	40562	40562	40562	40562
17	221952	221952	22690	199262	17686	17686	16531	16531	16531	16531	16531
19	389880	383724	16152	367572	12401	12401	11497	11497	11497	11497	11497
23	1024144	1001880	8821	993059	6683	6683	6392	6392	6392	6392	6392

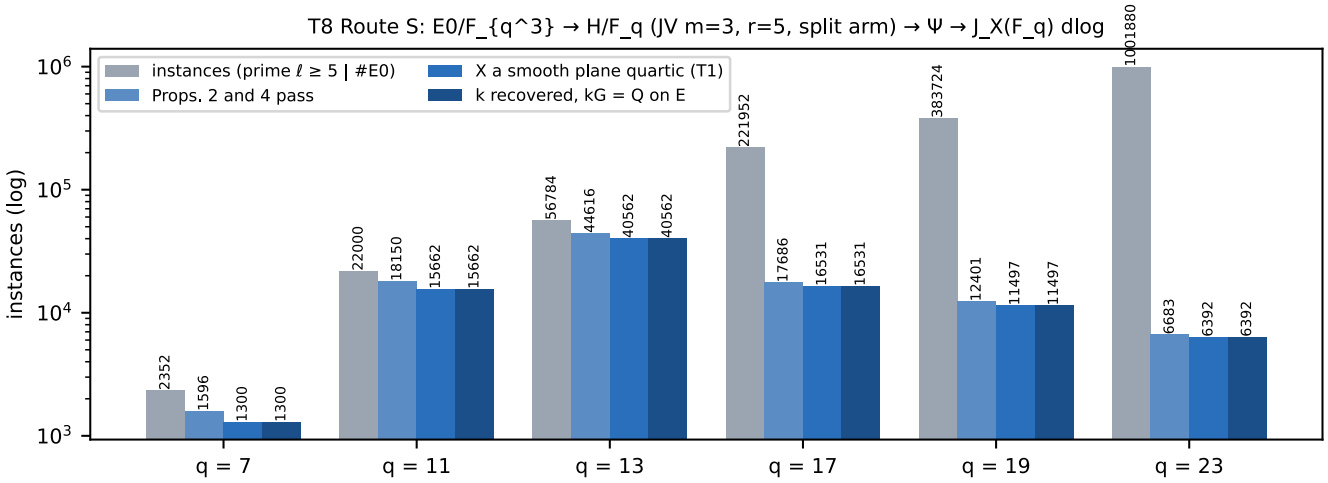


Figure 3: T8 per q (`data/route_s_q*.jsonl`, one row per instance with its skip reason).

Every DLP attempt whose quotient X is a smooth plane quartic ended with the planted k recovered and verified on E0; the others are skips with the recorded reason "images lie on a conic: X hyperelliptic" (Smith's quotient is then a hyperelliptic Jacobian and there is no plane quartic for this route). Props. 2 and 4 pass per instance at 67.9% (q = 7), 82.5% (11), 78.6% (13), 77.9% (17), 76.8% (19) and 75.8% (23) (table), against 18.6% predicted by Smith's Theorem 2 for a random genus-3 curve. Not like for like: every H here has cycle type (3,3,1,1) and three rational tractable subgroups, where a random curve has about one; at q ≥ 17 the instances are the first in order, not a sample. The DLP stage is brute force on canonical keys; nothing here is an index calculus or a cost comparison with rho (that is W2b).

5. T7: cost of ψ per divisor

fit of $\log_2$ y on $\log_2$ q (12 points)	slope	95% interval	intercept
F_q multiplications	0.099	[0.064, 0.133]	16.69
priced (inversions at measured time ratio)	0.107	[0.079, 0.135]	17.185
wall time (ns)	0.106	[0.087, 0.124]	16.978

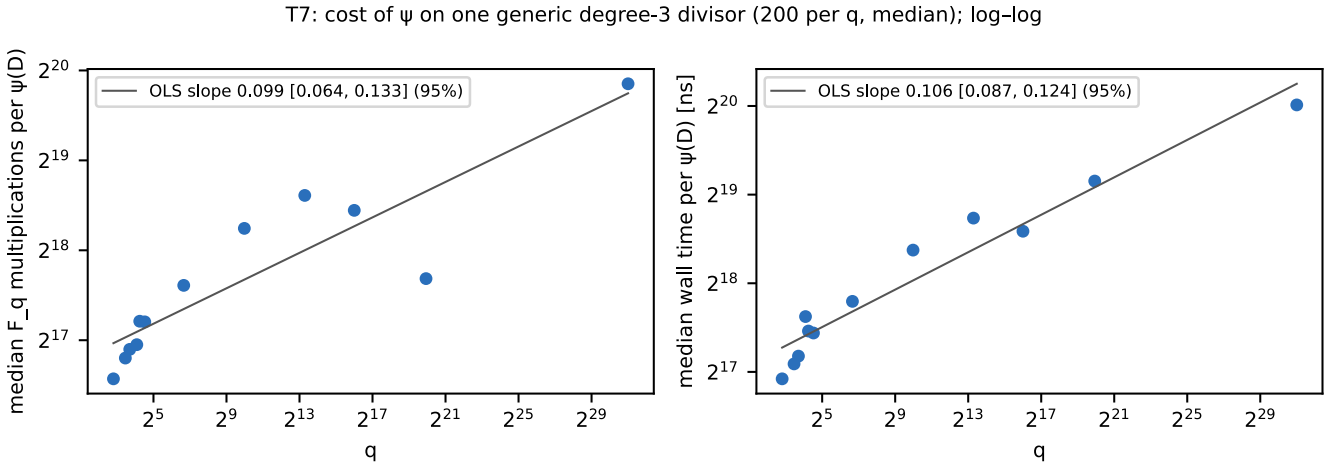


Figure 4: T7: median over 200 generic degree-3 divisors per q, q ∈ {7, ..., 23, 101, 1009, 10007, 65537, 1000003, $2^{31}-1$} (`data/cost_q*.jsonl`, fit `data/cost_fit.json`).

The slope is far below 1: consistent with the polylog model the protocol expected; the fit does not establish a polylog law. Unpriced in these counts: factoring u over F_q (`smith_prop24::factor_squarefree`), root finding in $F_{q^{2j}}$, Frobenius-orbit grouping and closed-point normalisation (counted only through their field operations), sorting and hashing. Wall times are from a shared host (load average above 80 during the runs), single thread, nice 10, not pinned; counts are unaffected by load.

6. Deviations from the frozen protocol (recorded; the protocol was not edited)

- JSONL rows are per curve (with per-check tallies) and per Route S instance, not one row per check instance.
- T1(f) at k = 1 only, on the degree-8 scan; T9(ii) for the degree-6 engine not run; T7 wall times not on pinned cores.
- T8 at q ≥ 17 ran under the 1 h cap; "not reached" counts the instances left, and the rows run are the first in the construction's order (lowest census index of r_0 first), not a uniform sample. q = 7, 11, 13 cover every instance (q = 13 was rerun with a 2 h cap after its first run hit 1 h at 53 173 of 56 784).
- A defect found during this study was fixed before the recorded runs: `Fld::sqrt`'s non-residue search did not terminate in some fields (e.g. q = 17, modulus $x^2 + 3x + 4$); it now enumerates field elements from z.

7. Files

`src/cryptanalysis/smith_isogeny.rs` (ψ, quartic, both engines' bridge, checks), `src/cryptanalysis/smith_route_s.rs` (T8), `examples/smith_isogeny.rs` (driver: `verify`, `deg8`, `route-s`, `cost`, `cost-fit`, `summarize`), `data/`, `reports/plot.py`, `reports/diagram.py`. Depends on W1 (PR #2227).